

Cracking the Code: Predicting the Mysteries of a Brown Dwarf

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ABSTRACT

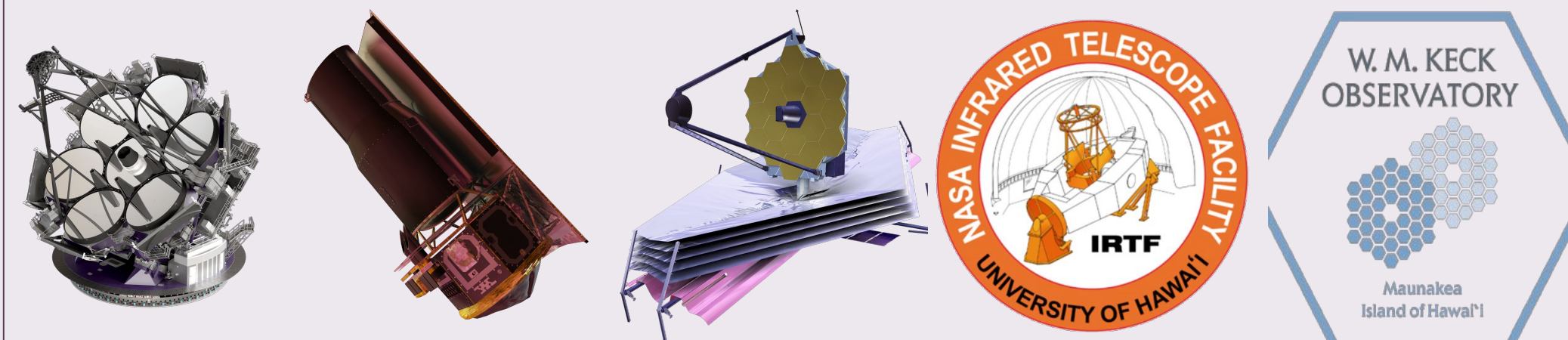
Our project involves using data from space and ground-based telescopes to analyze brown dwarfs. We compare these observations to computer models to **predict the age, mass, temperature, and gravity of the brown dwarfs**. This research helps us explore the relevance and limitations of these models in astrophysics. Our ultimate goal is to improve the theoretical and observational methods used in studying brown dwarfs.

BACKGROUND

A **brown dwarf is a celestial body heavier than a planet but isn't massive enough to be a star**. This lack of sufficient mass prevents self-sustaining nuclear fusion and causes brown dwarf cooling with age. As a result, its atmospheric chemical composition changes over time as it cools and dims, and we use those recorded chemical compositions to improve our knowledge on brown dwarfs^[1].

OBJECTIVES

- Analyse an astronomical data using astrophysical models.
- Predict physical parameters of the brown dwarf.
- Measure the accuracy of these models by comparing them with observations.



DATA

We received spectra—the brightness of an object at specific wavelengths of light — of the brown dwarf (0415-0935, class: T8) from the following telescopes^[2]:

- ★ James Webb Space Telescope
- ★ Keck Observatory
- ★ NASA Infrared Telescope Facility
- ★ Spitzer Space Telescope
- ★ Giant Magellan Telescope

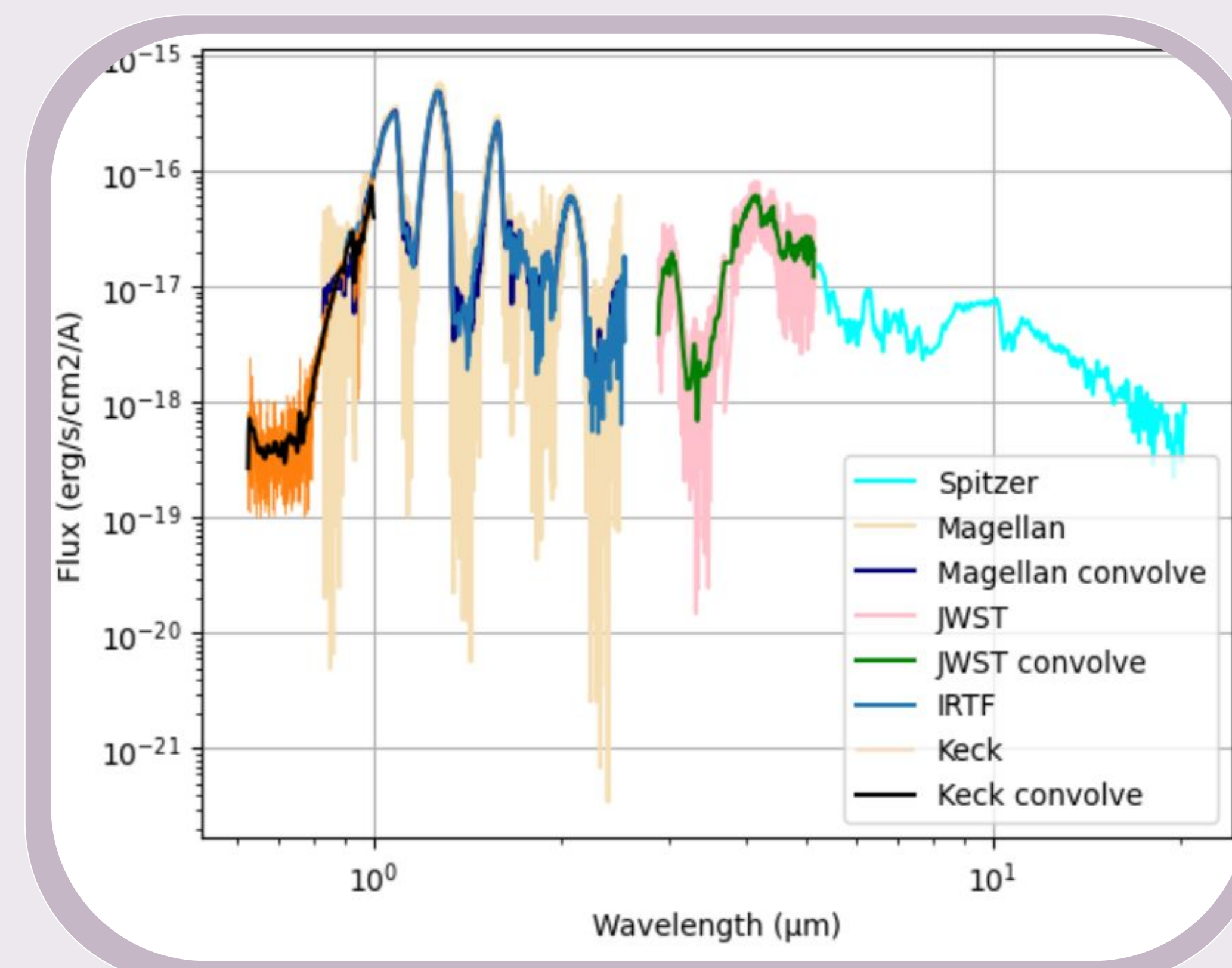


Fig: 0415-0935 Spectral Energy Distribution

	Temperature (K)	Gravity (logg)
Spitzer	700	4.6
Keck	390	4.1
JWST	680	5
IRTF	760	4.3
Magellan	740	4.3

METHODOLOGY

We constructed a spectral energy distribution (SED) using spectra from space and ground-based telescopes, inputting them into a Python notebook. To predict the dwarf's characteristics, we used ATMO2020^[3] atmospheric models, which are based on brown dwarf chemistry and physics. With SEDA^[4], **we selected the five best fit models for each spectrum** and took the median of the temperature and gravity predictions from these models. Using these two parameters, **we predicted mass, age, radius, and luminosity with evolutionary models**^[3].

RESULTS

Parameter	Value	Error
Temperature (K)	700	49
Gravity (logg)	4.3	0.3
Mass (M _{jup})	9.9	7.2
Age (Gyr)	0.2	0.4
Radius (R _{jup})	1.1	0.1
Luminosity (logL)	-5.55	0.04

Table: Predictions of parameters based on Teff and logG using evolutionary models

Table: Temperature and Gravity from the 5 Telescopes

DISCUSSION

We obtain the the brown dwarf was **10 times more massive than jupiter, but 100 times less massive than the Sun**. The temperature of the brown dwarf is only a few hundred kelvin above the average temperature for baking pizza! Brown dwarfs truly are not as crazy as one may think!

There were some limitations in wavelength, but the models were accurate. Ultimately, brown dwarfs are hard to study due to their nature but fortunately because of advancements in astrophysics we were able to document these phenomenons.

ACKNOWLEDGEMENTS

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REFERENCES

- [1] - Kirkpatrick, J.D., 2005, ARA&A, 43, 195
- [2] - Alejandro Merchan et al. 2024, in prep.
- [3] - Phillips, M.W., et al. 2020, A&A, 637, A38
- [4] - Suárez, G., et al. 2024, in prep.